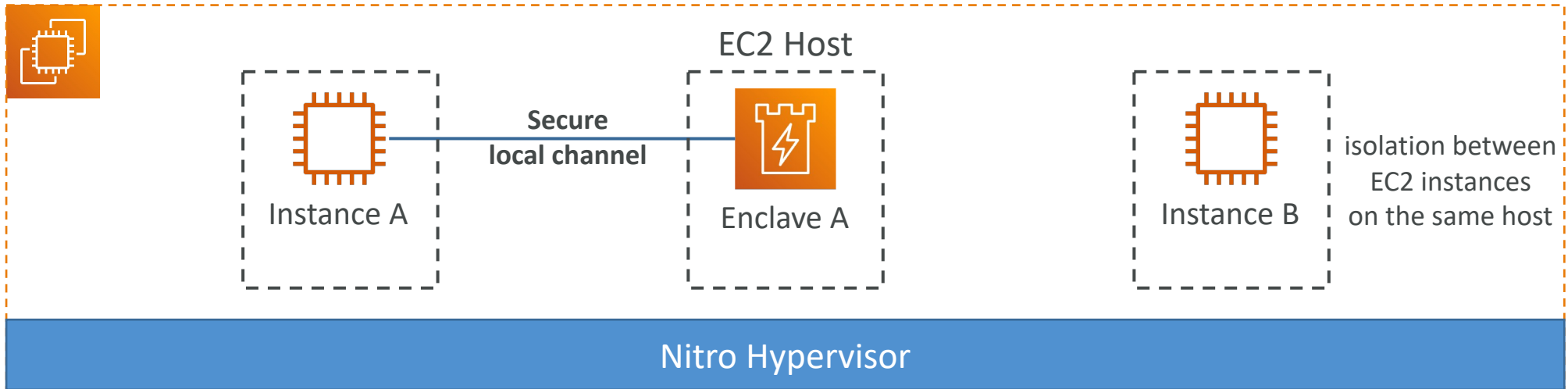
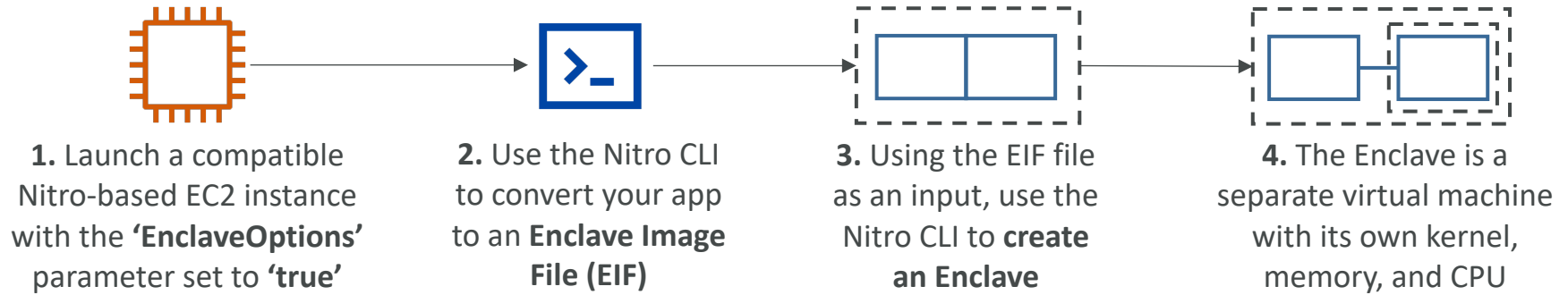


AWS Nitro Enclaves



Other Services

Quick overview of other services that might have questions on at the exam

Amazon Simple Email Service (Amazon SES)



- Fully managed service to send emails securely, globally and at scale
- Allows inbound/outbound emails
- Reputation dashboard, performance insights, anti-spam feedback
- Provides statistics such as email deliveries, bounces, feedback loop results, email open
- Supports DomainKeys Identified Mail (DKIM) and Sender Policy Framework (SPF)
- Flexible IP deployment: shared, dedicated, and customer-owned IPs
- Send emails using your application using AWS Console, APIs, or SMTP
- Use cases: transactional, marketing and bulk email communications



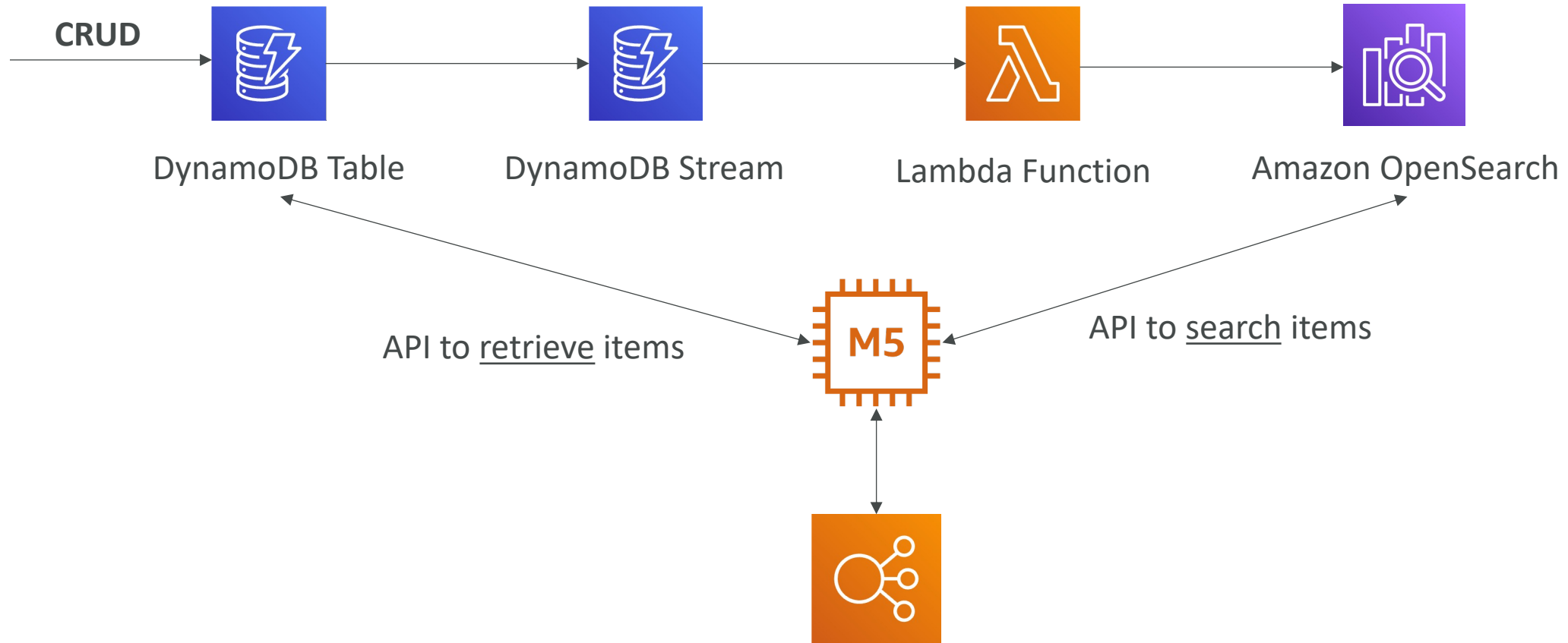


Amazon OpenSearch Service

- *Amazon OpenSearch is successor to Amazon ElasticSearch*
- In DynamoDB, queries only exist by primary key or indexes...
- **With OpenSearch, you can search any field, even partially matches**
- It's common to use OpenSearch as a complement to another database
- Two modes: managed cluster or serverless cluster
- Does *not* natively support SQL (can be enabled via a plugin)
- Ingestion from Kinesis Data Firehose, AWS IoT, and CloudWatch Logs
- Security through Cognito & IAM, KMS encryption, TLS
- Comes with OpenSearch Dashboards (visualization)

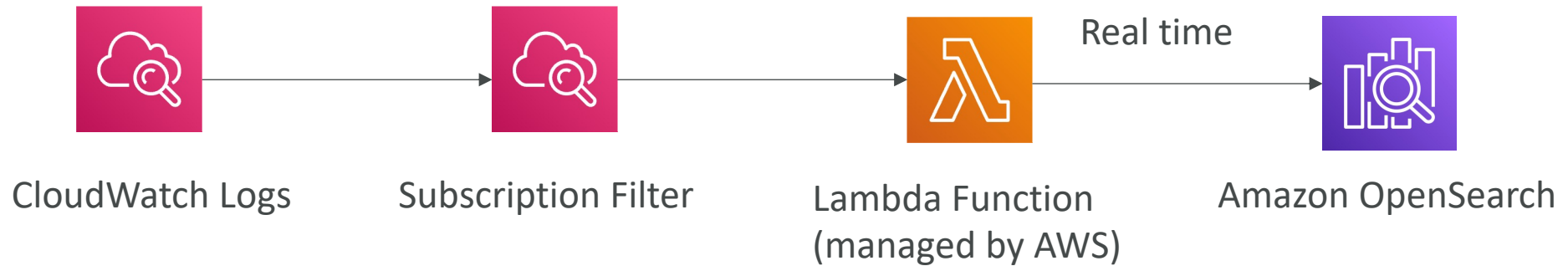
OpenSearch patterns

DynamoDB



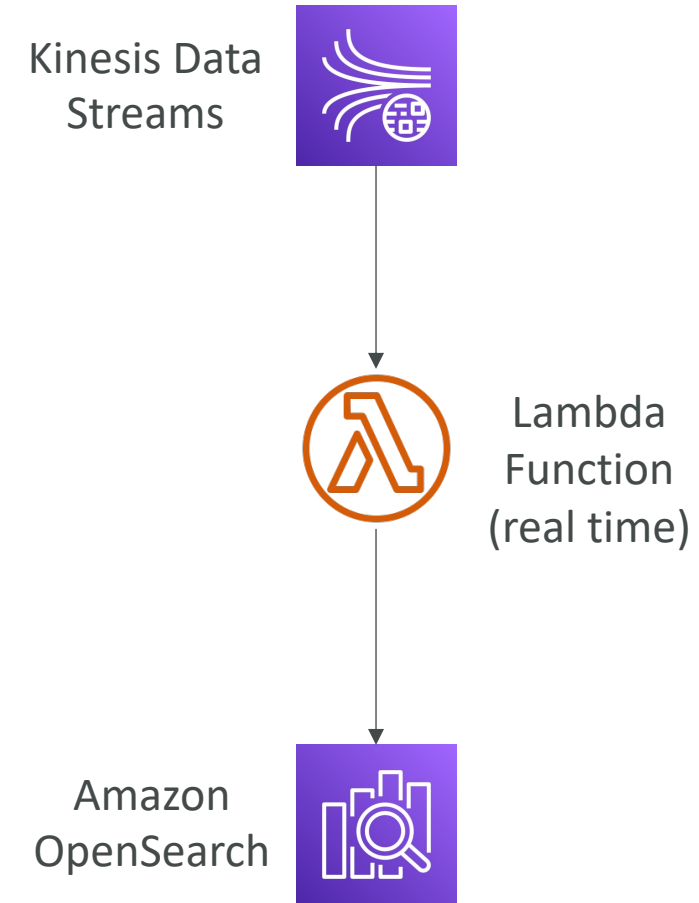
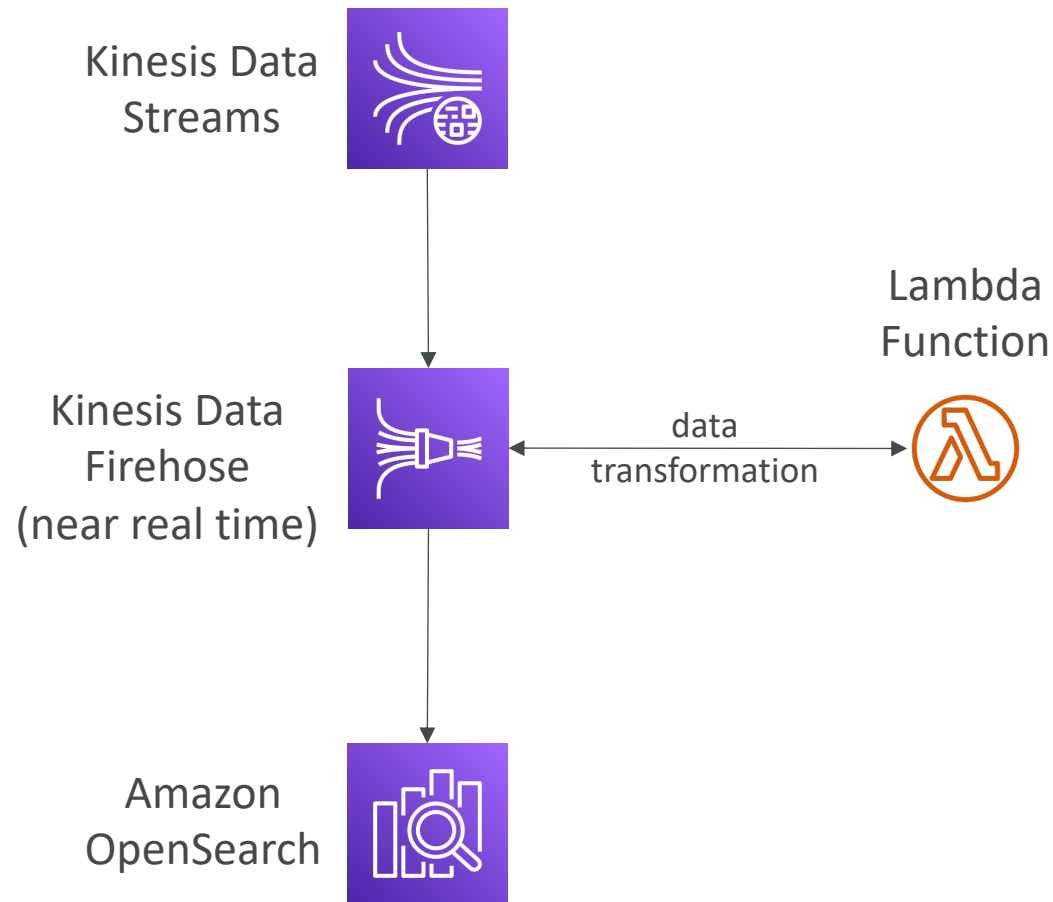
OpenSearch patterns

CloudWatch Logs



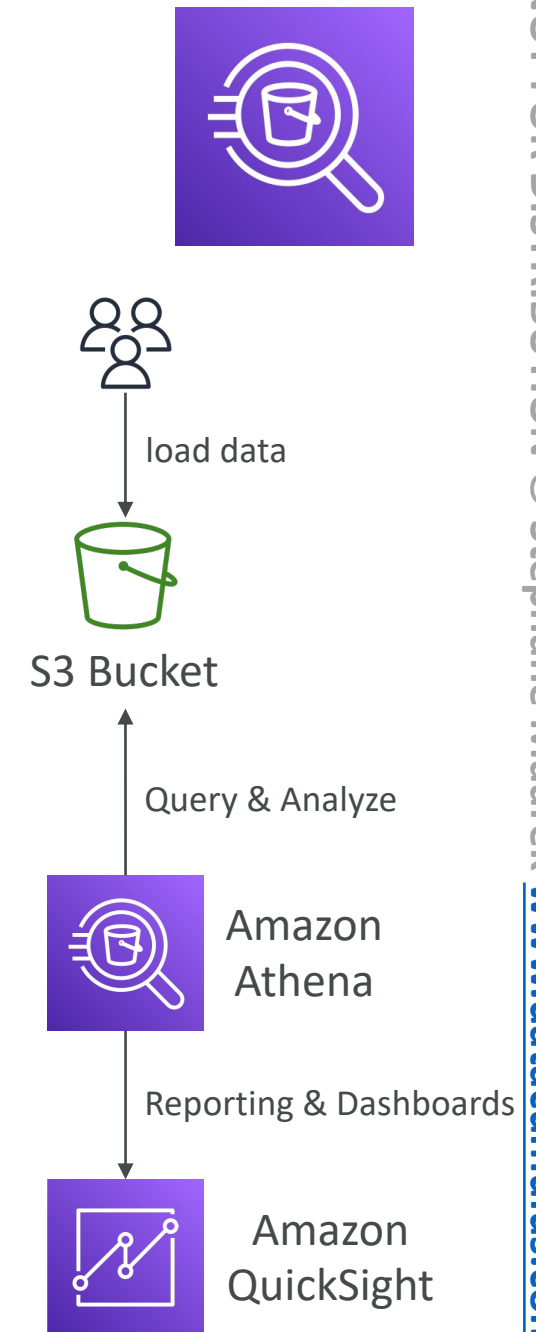
OpenSearch patterns

Kinesis Data Streams & Kinesis Data Firehose



Amazon Athena

- **Serverless** query service to analyze data stored in Amazon S3
- Uses standard SQL language to query the files (built on Presto)
- Supports CSV, JSON, ORC, Avro, and Parquet
- Pricing: \$5.00 per TB of data scanned
- Commonly used with Amazon Quicksight for reporting/dashboards
- **Use cases:** Business intelligence / analytics / reporting, analyze & query VPC Flow Logs, ELB Logs, **CloudTrail trails**, etc...
- **Exam Tip:** analyze data in S3 using serverless SQL, use Athena

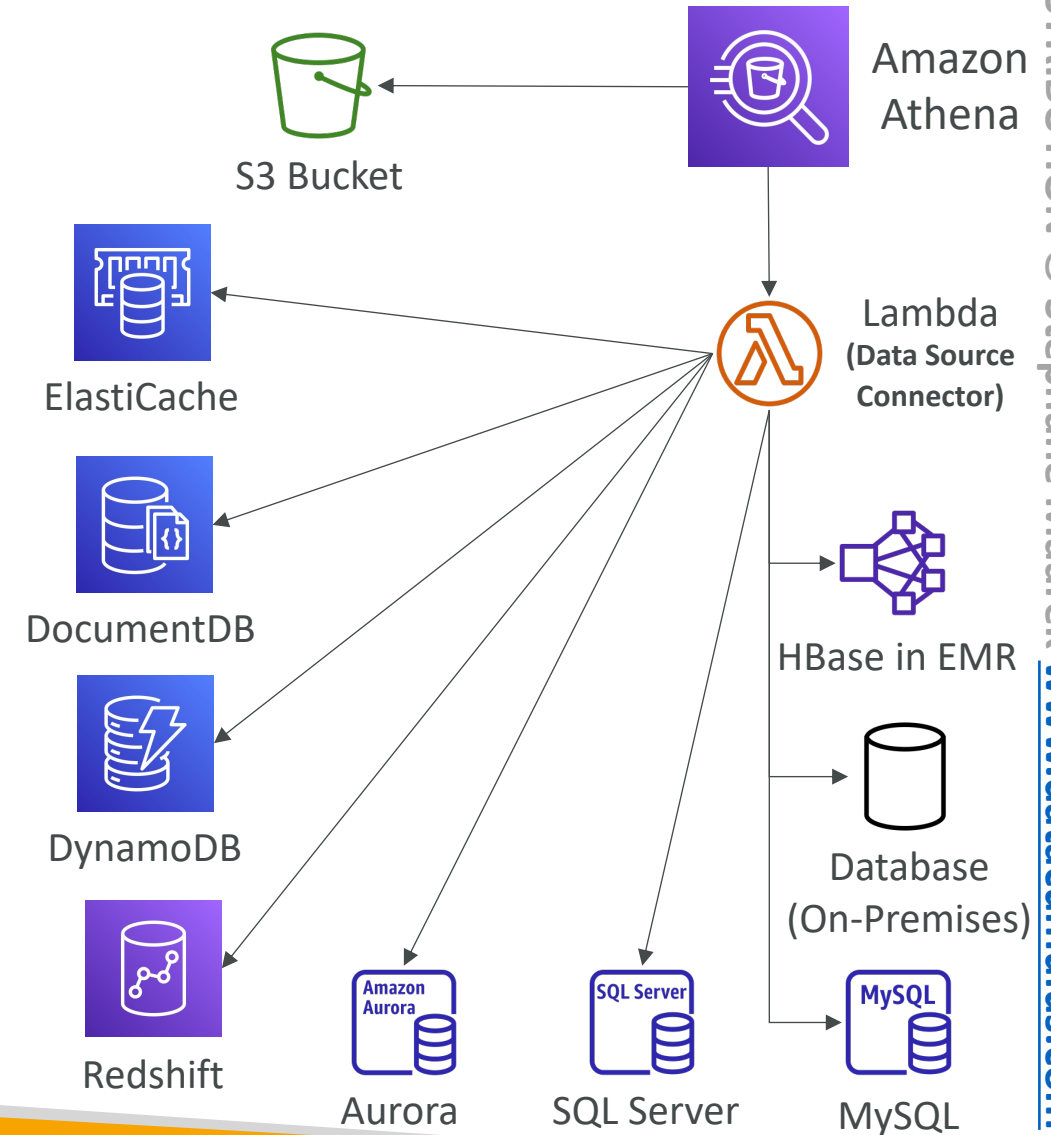


Amazon Athena – Performance Improvement

- Use **columnar data** for cost-savings (less scan)
 - Apache Parquet or ORC is recommended
 - Huge performance improvement
 - Use Glue to convert your data to Parquet or ORC
- **Compress data** for smaller retrievals (bzip2, gzip, lz4, snappy, zlip, zstd...)
- **Partition** datasets in S3 for easy querying on virtual columns
 - s3://yourBucket/pathToTable
 - /<PARTITION_COLUMN_NAME>=<VALUE>
 - /<PARTITION_COLUMN_NAME>=<VALUE>
 - /<PARTITION_COLUMN_NAME>=<VALUE>
 - /etc...
 - Example: s3://athena-examples/flight/parquet/year=1991/month=1/day=1/
- Use **larger files** (> 128 MB) to minimize overhead

Amazon Athena – Federated Query

- Allows you to run SQL queries across data stored in relational, non-relational, object, and custom data sources (AWS or on-premises)
- Uses Data Source Connectors that run on AWS Lambda to run Federated Queries (e.g., CloudWatch Logs, DynamoDB, RDS, ...)
- Store the results back in Amazon S3

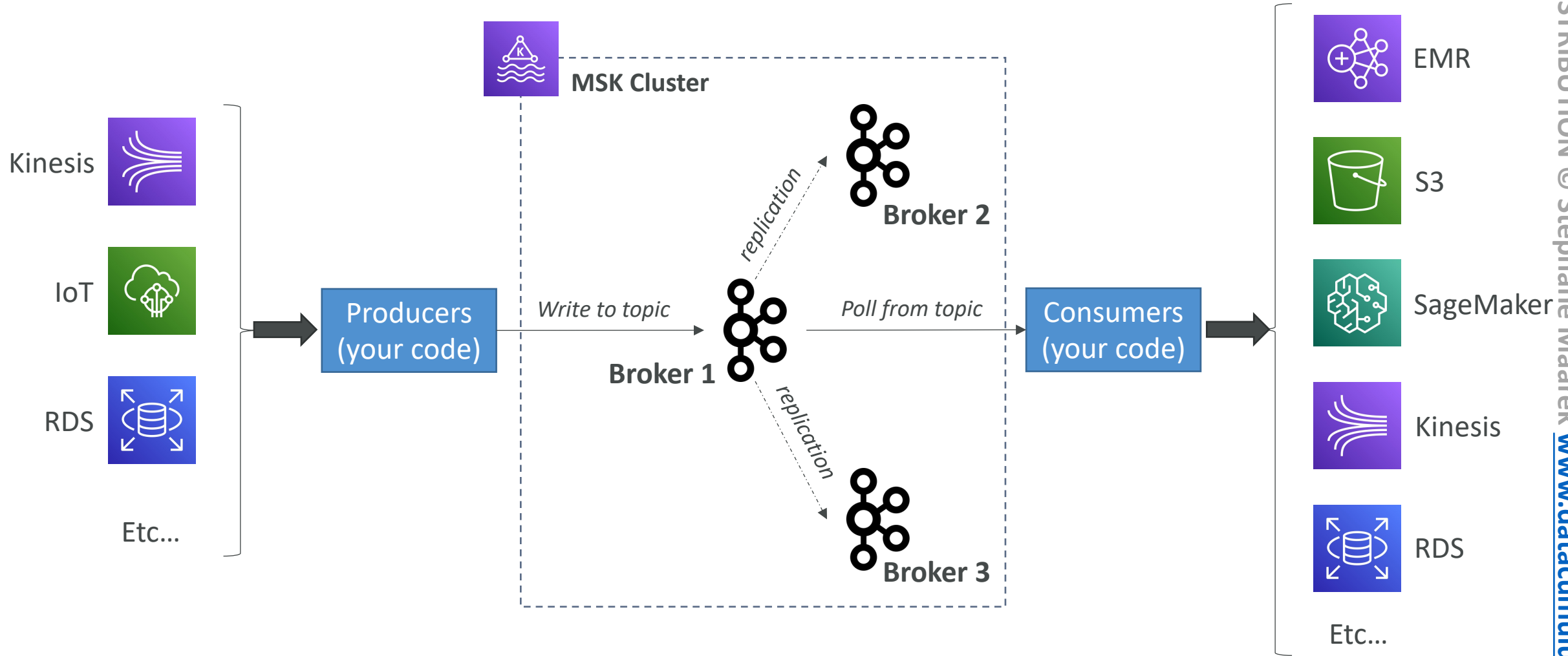


Amazon Managed Streaming for Apache K (Amazon MSK)



- Alternative to Amazon Kinesis
- Fully managed Apache Kafka on AWS
 - Allow you to create, update, delete clusters
 - MSK creates & manages Kafka brokers nodes & Zookeeper nodes for you
 - Deploy the MSK cluster in your VPC, multi-AZ (up to 3 for HA)
 - Automatic recovery from common Apache Kafka failures
 - Data is stored on EBS volumes **for as long as you want**
- **MSK Serverless**
 - Run Apache Kafka on MSK without managing the capacity
 - MSK automatically provisions resources and scales compute & storage

Apache Kafka at a high level



Kinesis Data Streams vs. Amazon MSK



Kinesis Data Streams

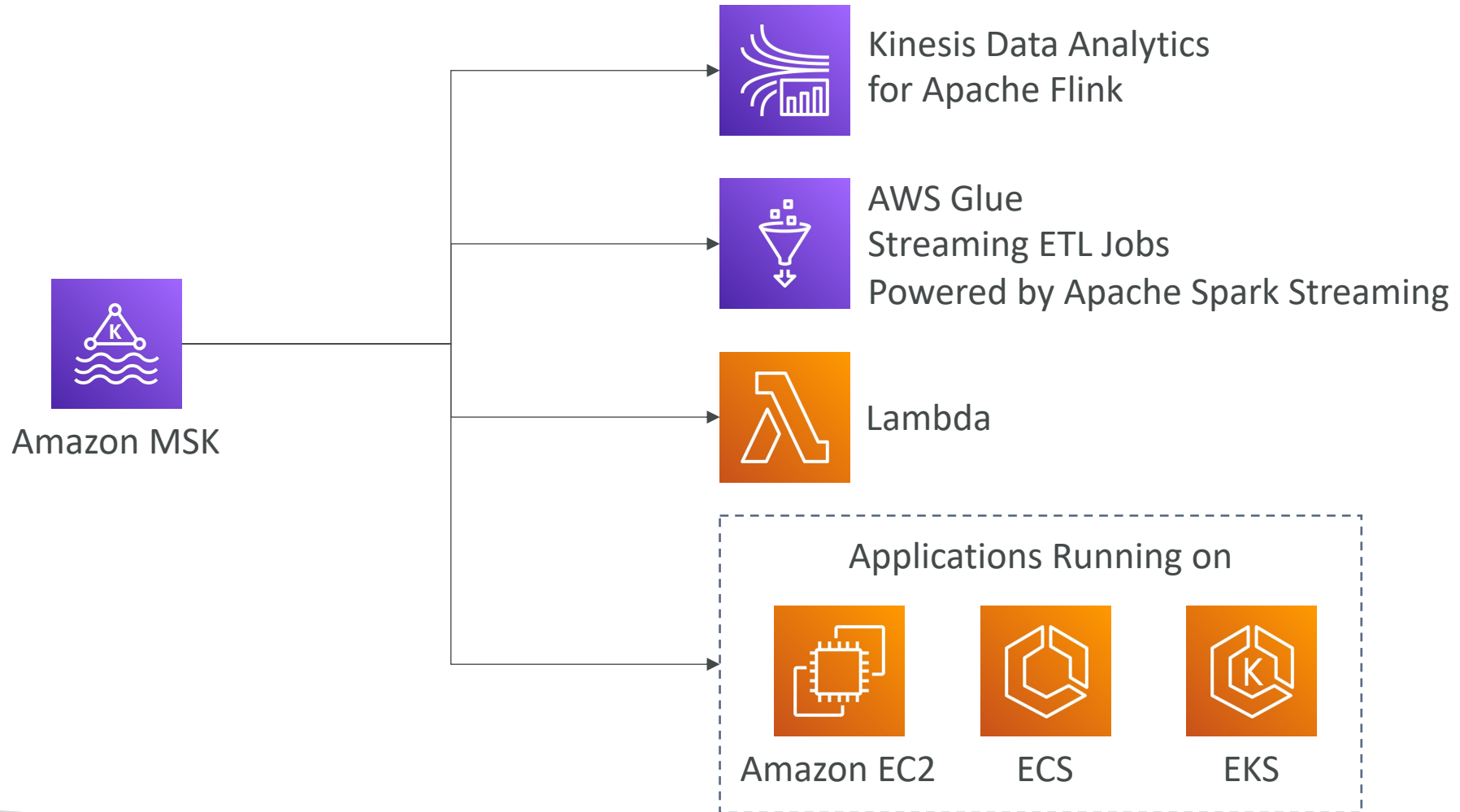
- 1 MB message size limit
- Data Streams with Shards
- Shard Splitting & Merging
- TLS In-flight encryption
- KMS at-rest encryption



Amazon MSK

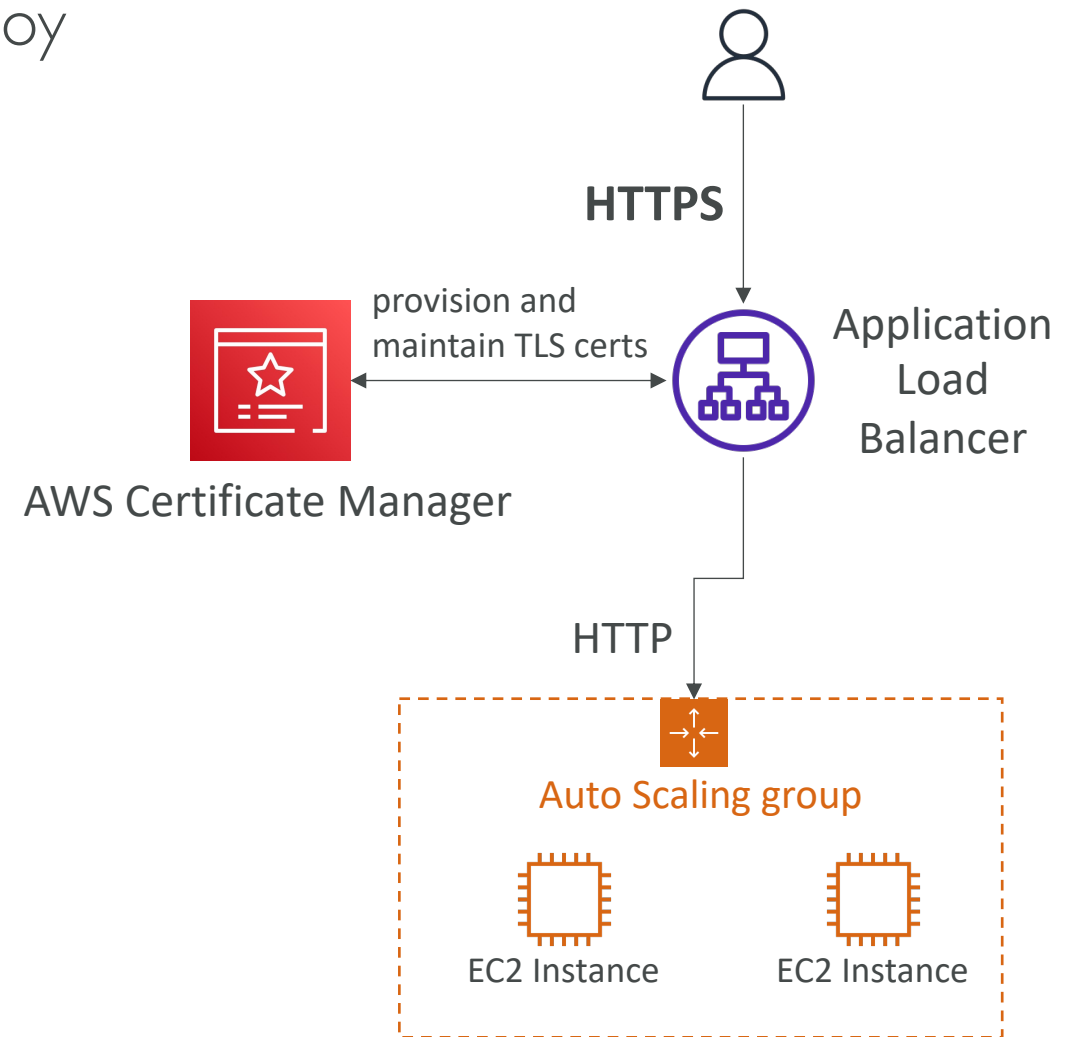
- 1 MB default, configure for higher (ex: 10MB)
- Kafka Topics with Partitions
- Can only add partitions to a topic
- PLAINTEXT or TLS In-flight Encryption
- KMS at-rest encryption

Amazon MSK Consumers



AWS Certificate Manager (ACM)

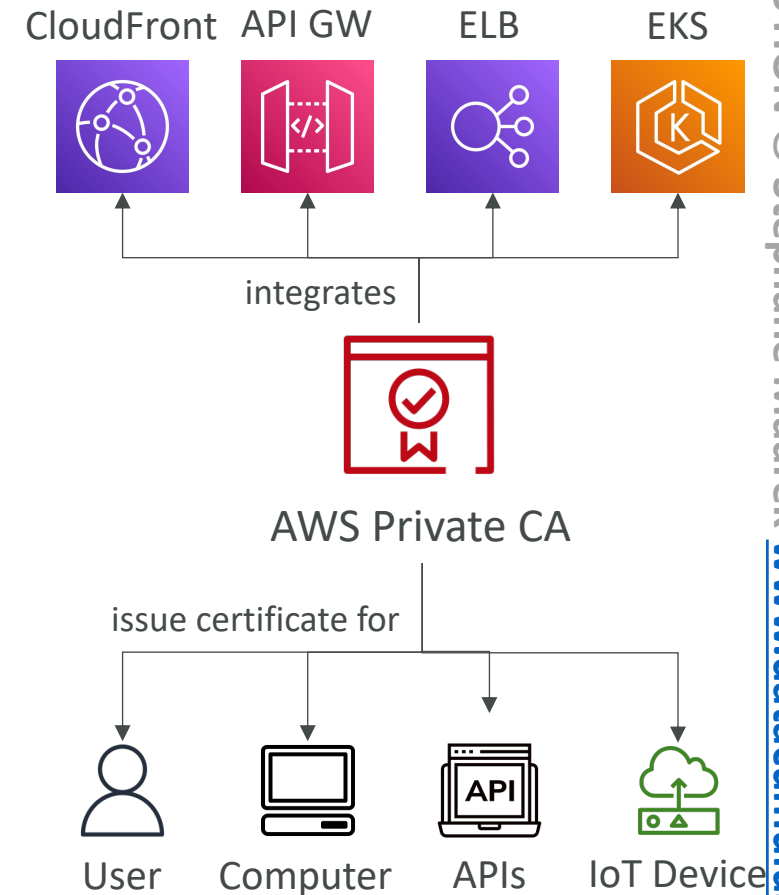
- Let's you easily provision, manage, and deploy **SSL/TLS Certificates**
- Used to provide in-flight encryption for websites (HTTPS)
- Supports both public and private TLS certificates
- Free of charge for public TLS certificates
- Automatic TLS certificate renewal
- Integrations with (load TLS certificates on)
 - Elastic Load Balancers
 - CloudFront Distributions
 - APIs on API Gateway



AWS Private Certificate Authority (CA)



- Managed service allows you to create private Certificate Authorities (CA), including root and subordinates CAs
- Can issue and deploy end-entity X.509 certificates
- Certificates are trusted only by your Organization (not the public Internet)
- Works for AWS services that are integrated with ACM
- Use cases:
 - Encrypted TLS communication, Cryptographically signing code
 - Authenticate users, computers, API endpoints, and IoT devices
 - Enterprise customers building a Public Key Infrastructure (PKI)



AWS Macie

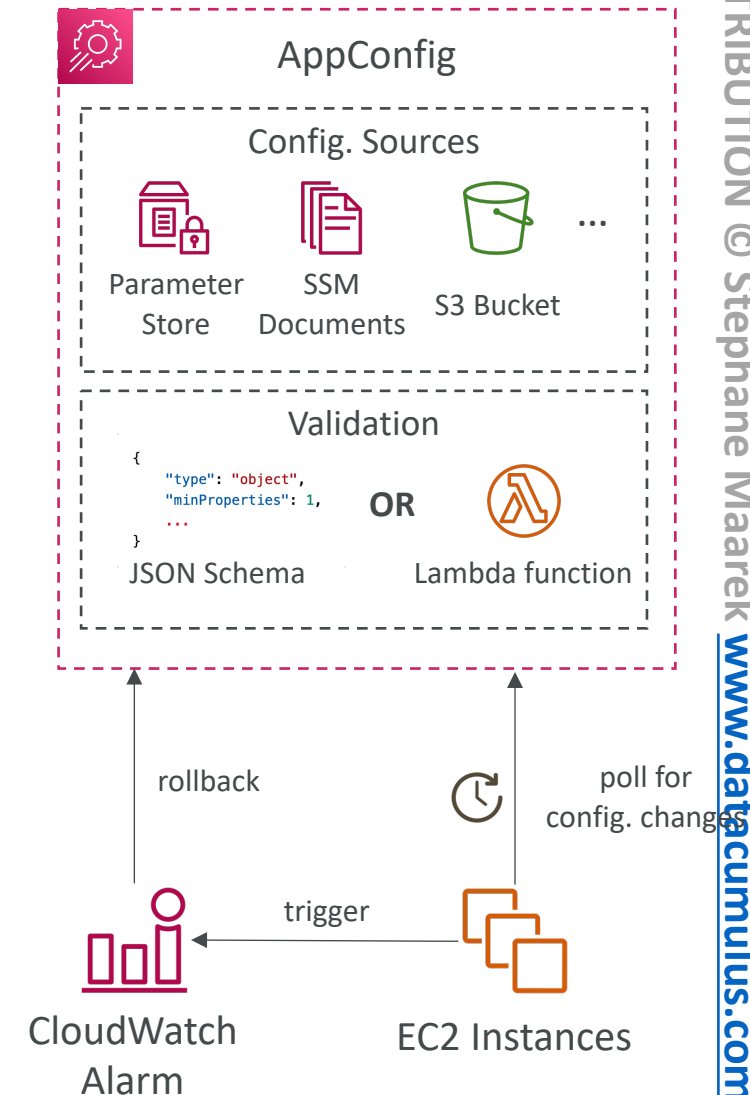


- Amazon Macie is a fully managed data security and data privacy service that uses **machine learning** and **pattern matching** to **discover and protect your sensitive data in AWS**.
- Macie helps identify and alert you to **sensitive data**, such as **personally identifiable information (PII)**



AWS AppConfig

- Configure, validate, and deploy dynamic configurations
- Deploy dynamic configuration changes to your applications independently of any code deployments
 - You don't need to restart the application
- Feature flags, application tuning, allow/block listing...
- Use with apps on EC2 instances, Lambda, ECS, EKS...
- Gradually deploy the configuration changes and rollback if issues occur
- Validate configuration changes before deployment using:
 - **JSON Schema** (syntactic check) or
 - **Lambda Function** – run code to perform validation (semantic check)



Exam Review & Tips

State of learning checkpoint

- Let's look how far we've gone on our learning journey
- <https://aws.amazon.com/certification/certified-developer-associate/>

Practice makes perfect

- If you're new to AWS, take a bit of AWS practice thanks to this course before rushing to the exam
- The exam recommends you to have one or more years of hands-on developing and maintaining an AWS based applications
- Practice makes perfect!
- If you feel overwhelmed by the amount of knowledge you just learned, just go through it one more time

Ideas for practicing....!

- Take one of your existing applications
- Try deploying it manually on EC2
- Try deploying it on Elastic Beanstalk and have it scale
- Try creating a CI/CD pipeline for it
- Try decoupling components using SQS / SNS
- If possible, try running it on AWS Lambda & friends
- Write automation scripts using the CLI / SDK
 - Idea 1: Shut down EC2 instances at night / start in the morning
 - Idea 2: Automate snapshots of EBS volumes at night
 - Idea 3: List all under-utilized EC2 instances (CPU Utilization < 10%)

Proceed by elimination

- Most questions are going to be scenario based
- For all the questions, rule out answers that you know for sure are wrong
- For the remaining answers, understand which one makes the most sense
- There are very few trick questions
- Don't over-think it
- If a solution seems feasible but highly complicated, it's probably wrong